Jiwan Chung

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Google Scholar

Research Profile

Ph.D. candidate in Artificial Intelligence at Yonsei University working on multimodal intelligence, with a focus on reasoning, grounding, and agentic behavior. My research combines model design with diagnostic evaluation to improve how multimodal systems reason over visual evidence and make decisions across steps.

Research Areas

- Multimodal reasoning
- Grounded interaction and vision-language alignment
- Evaluation and diagnosis of AI systems
- Robustness and interpretability of multimodal models

Education

2023-26*	**Ph.D.**, Artificial Intelligence, Yonsei University *Advisor: Seon Joo Kim*; GPA: 4.3 / 4.3
2019-23	**M.S.**, Computer Engineering, Seoul National University *Advisor: Gunhee Kim*; GPA: 3.75 / 4.3
2014-18	**B.Sc.**, Computer Science and Philosophy, Yonsei University *Double Major*; GPA: 3.84 / 4.30

Research Experience

2025 Summer	**Microsoft Research**, AI Frontiers Conducted research on diagnosing multimodal reasoning capabilities in MLLMs (MathLens; arXiv 2025), with the collaboration later continuing into on process-level evaluation of web agents (WebStep; 2026).	Research Scientist Intern
2024 Spring	**LG AI Research**, AML Conducted research on extending multimodal LLMs beyond language-only outputs, including embodied VLA, image generation, and mathematical reasoning (DIST2Loss; ICLR 2026).	Research Scientist Intern
2023 Fall	**Naver**, Foundational Research Contributed to HyperCLOVA X LLM development and conducted preliminary exploration on omnimodal language modeling.	Research Intern

*Expected.

Selected Research

2026	**Where Did It Go Wrong? Process-Level Evaluation of Web Agents with Semantic State Tracking** *Preprint*. Developed a controlled benchmark for diagnosing process-level failures in web agents, enabling fine-grained analysis of exploration, execution, and decision-making behaviors.
2025	**v1: Learning to Point Visual Tokens for Multimodal Grounded Reasoning** *Preprint*. Developed visual pointing mechanisms to improve grounded reasoning in multimodal LLMs, enhancing their ability to align visual evidence with multi-step reasoning.
2026	**Teaching Metric Distance to Discrete Autoregressive Language Models** *ICLR 2026*. Introduced a training objective that teaches discrete autoregressive language models to represent metric distance, improving their ability to capture spatial and relational structure for embodied AI applications.
2025	**Are Any-to-Any Models More Consistent Across Modality Transfers Than Specialists?** *ACL 2025*. Developed a benchmark for evaluating consistency across multimodal transfer pathways, enabling fine-grained diagnosis of cross-modal behavior in any-to-any multimodal LLMs.

Conference Papers

G Google Scholar
† → Equal contribution

. *Global Geometry Is Not Enough for Vision Representations*
Jiwan Chung & Seon Joo Kim
ICML 2026

. *Teaching Metric Distance to Discrete Autoregressive Language Models*
Jiwan Chung, Saejin Kim, Yongrae Jo, Jaewoo Park, Dongjun Min, Youngjae Yu
ICLR 2026

. *Are Any-to-Any Models More Consistent Across Modality Transfers Than Specialists?*
Jiwan Chung, Janghan Yoon, Junhyeong Park, Sangeyl Lee, Joowon Yang, Sooyeon Park, Youngjae Yu
ACL 2025

. *MASS: Overcoming Language Bias in Image-Text Matching*
Jiwan Chung, Seungwon Lim, Sangkyu Lee, Youngjae Yu
AAAI 2025

. *Speaking Beyond Language: A Large-Scale Multimodal Dataset for Learning Nonverbal Cues from Video-Grounded Dialogues*

Jiwan Chung†, Youngmin Kim†, Jisoo Kim, Sunghyun Lee, Sangkyu Lee, Junhyeok Kim, Cheoljong Yang, Youngjae Yu
ACL **2025**

- ***Can visual language models resolve textual ambiguity with visual cues? Let visual puns tell you!***
  Jiwan Chung, Seungwon Lim, Jaehyun Jeon, Seungbeen Lee, Yu Youngjae
  EMNLP **2024**
- ***Selective Vision is the Challenge for Visual Reasoning: A Benchmark for Visual Argument Understanding***
  Jiwan Chung†, Sungjae Lee†, Minseo Kim, Seungju Han, Ashkan Yousefpour, Jack Hessel, Youngjae Yu
  EMNLP **2024**
- ***Towards Visual Text Design Transfer Across Languages***
  Jiwan Chung†, Yejin Choi†, Sumin Shim, Giyeong Oh, Youngjae Yu
  NeurIPS D&B **2024**
- ***Fusing pre-trained language models with multimodal prompts through reinforcement learning***
  Jiwan Chung†, Youngjae Yu†, Heeseung Yun, Jack Hessel, Jae Sung Park, Ximing Lu, Rowan Zellers, Prithviraj Ammanabrolu, Ronan Le Bras, Gunhee Kim
  CVPR **2023**
- ***Long Story Short: a Summarize-then-Search Method for Long Video Question Answering***
  Jiwan Chung, Youngjae Yu
  BMVC **2023**
- ***VLIS: Unimodal Language Models Guide Multimodal Language Generation***
  Jiwan Chung, Youngjae Yu
  EMNLP **2023**
- ***Acav100m: Automatic curation of large-scale datasets for audio-visual video representation learning***
  Jiwan Chung†, Sangho Lee†, Youngjae Yu, Gunhee Kim, Thomas Breuel, Gal Chechik, Yale Song
  ICCV **2021**
- ***Transitional adaptation of pretrained models for visual storytelling***
  Jiwan Chung†, Youngjae Yu†, Heeseung Yun, Jongseok Kim, Gunhee Kim
  CVPR **2021**

- ***Explain with Visual Keypoints Like a Real Mentor! A Benchmark for Multimodal Solution Explanation***
  Jaewoo Park, Jungyang Park, Dongju Jang, **Jiwan Chung**, Byungwoo Yoo, Jaewoo Shin, Seonjoon Park, Taehyeong Kim, Youngjae Yu
  AAAI **2026**
- ***GuideDog: A Real-World Egocentric Multimodal Dataset for Blind and Low-Vision Accessibility-Aware Guidance***
  Junhyeok Kim, Jaewoo Park, Junhee Park, Sangeyl Lee, **Jiwan Chung**, Jisung Kim, Ji Hoon Joung, Youngjae Yu
  ACL **2026**

- *CANVAS: Commonsense-Aware Navigation System for Intuitive Human-Robot Interaction*
  Suhwan Choi, Yongjun Cho, Minchan Kim, Jaeyoon Jung, Myunchul Joe, Yubeen Park, Minseo Kim, Sungwoong Kim, Sungjae Lee, Hwiseong Park, **Jiwan Chung**, Youngjae Yu
  ICRA **2025**
- *Do LLMs Have Distinct and Consistent Personality? TRAIT: Personality Testset designed for LLMs with Psychometrics*
  Seungbeen Lee, Seungwon Lim, Seungju Han, Giyeong Oh, Hyungjoo Chae, **Jiwan Chung**, Minju Kim, Beong-woo Kwak, Yeonsoo Lee, Dongha Lee, Jinyoung Yeo, Youngjae Yu
  NAACL Findings **2025**
- *EgoSpeak: Learning When to Speak for Egocentric Conversational Agents in the Wild*
  Junhyeok Kim, Minsoo Kim, **Jiwan Chung**, Jungbin Cho, Jisoo Kim, Sungwoong Kim, Gyeongbo Sim, Youngjae Yu
  NAACL Findings **2025**
- *VAGUE: Visual Contexts Clarify Ambiguous Expressions*
  Heejeong Nam, Jinwoo Ahn, Keummin Ka, **Jiwan Chung**, Youngjae Yu
  ICCV **2025**
- *VisEscape: A Benchmark for Evaluating Exploration-driven Decision-making in Virtual Escape Rooms*
  Seungwon Lim, Sungwoong Kim, Jihwan Yu, Sungjae Lee, **Jiwan Chung**, Youngjae Yu
  EMNLP **2025**
- *Language models as compilers: Simulating pseudocode execution improves algorithmic reasoning in language models*
  Hyungjoo Chae, Yeonghyeon Kim, Seungone Kim, Kai Tzu-iunn Ong, Beong-woo Kwak, Moohyeon Kim, Seonghwan Kim, Taeyoon Kwon, **Jiwan Chung**, Youngjae Yu
  EMNLP **2024**
- *Reading Books is Great, But Not if You Are Driving! Visually Grounded Reasoning about Defeasible Commonsense Norms*
  Seungju Han, Junhyeok Kim, Jack Hessel, Liwei Jiang, **Jiwan Chung**, Yejin Son, Yejin Choi, Youngjae Yu
  EMNLP **2023**
- *Character grounding and re-identification in story of videos and text descriptions*
  Youngjae Yu, Jongseok Kim, Heeseung Yun, **Jiwan Chung**, Gunhee Kim
  ECCV **2020**

Preprints

- **Rethinking State Tracking in Recurrent Models Through Error Control Dynamics**
  Jiwan Chung, Heechan Choi & Seon Joo Kim
  arXiv **2026**
- **Where Did It Go Wrong? Process-Level Evaluation of Web Agents with Semantic State Tracking**
  Jiwan Chung, JiHuyk Byun, Vibhav Vineet & Seon Joo Kim
  Preprint **2026**
- **v1: Learning to Point Visual Tokens for Multimodal Grounded Reasoning**

Jiwan Chung, Junhyeok Kim, Siyeol Kim, Jaeyoung Lee, Min Soo Kim, Youngjae Yu
arXiv **2025**

. **What MLLMs Learn about When they Learn about Multimodal Reasoning: Perception, Reasoning, or their Integration?**
Jiwan Chung, Neel Joshi, Pratyusha Sharma, Youngjae Yu, Vibhav Vineet
arXiv **2025**

. **SEAL: Entangled White-box Watermarks on Low-Rank Adaptation**
Giyeong Oh, Saejin Kim, Woohyun Cho, Sangkyu Lee**, Jiwan Chung**, Dokyung Song, Youngjae Yu
arXiv **2025**

. **HyperCLOVA X Technical Report**
Kang Min Yoo, Jaegeun Han, Sookyo In, Heewon Jeon, Jisu Jeong, Jaewook Kang, Hyunwook Kim, Kyung-Min Kim, Munhyong Kim, Sungju Kim, HyperCLOVA X Team
arXiv **2024**

Awards & Honors

2025	AI Seoul Tech Graduate Scholarship
2020	DramaQA2020 Challenge (ECCV Workshop), 1st place
2019	LSMDC2019 Challenge (ICCV Workshop), 1st place

Academic Service

Reviewer

- NeurIPS, ICLR, ICML
- CVPR, ICCV, ECCV
- ACL, NAACL, EMNLP
- COLM

Last updated: May 19, 2026